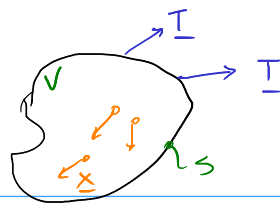


Ecuaciones de movimiento



Cantidad de movimiento lineal: $P_i = \int_V \rho v_i dV$

Resultante de los fuerzas externas: $F_i = \int_S T_i dS + \int_V x_i dV$

Fórmula de Cauchy: $T_i = n_j \tau_{ji}$

$\int_S T_i dS = \int_S n_j \tau_{ji} dS = \int_V \frac{\partial \tau_{ji}}{\partial x_j} dV$

$F_i = \int_V \left(\frac{\partial \tau_{ji}}{\partial x_j} + x_i \right) dV$

leyes de Newton: $\frac{D}{Dt}(P_i) = F_i$ $\frac{D}{Dt} \int_V A dV = \int_V \left(\frac{DA}{Dt} + A \frac{\partial v_j}{\partial x_j} \right) dV$

$\frac{D}{Dt}(P_i) = \frac{D}{Dt} \left(\int_V \rho v_i dV \right) = \int_V \left(\frac{D}{Dt}(\rho v_i) + \rho v_i \frac{\partial v_i}{\partial x_i} \right) dV$

$= \int_V \left(\frac{D\rho}{Dt} v_i + \rho \frac{Dv_i}{Dt} + \rho v_i \frac{\partial v_i}{\partial x_i} \right) dV$

Ecuación de continuidad:

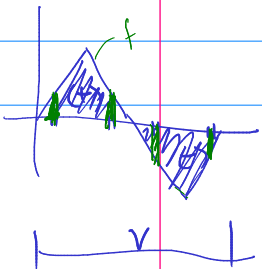
$\frac{D\rho}{Dt} + \rho \frac{\partial v_j}{\partial x_j} = 0$

$= \int_V v_i \left(\frac{D\rho}{Dt} + \rho \frac{\partial v_i}{\partial x_i} \right) dV + \int_V \rho \frac{Dv_i}{Dt} dV =$

$\int_V \rho \frac{Dv_i}{Dt} dV = \int_V \left(\frac{\partial \tau_{ji}}{\partial x_j} + x_i \right) dV$

$\int_V \left[\rho \frac{Dv_i}{Dt} - \left(\frac{\partial \tau_{ji}}{\partial x_j} + x_i \right) \right] dV = 0$

$f \Rightarrow f=0$



$$\rho \cdot \frac{D}{Dt} v_i = \frac{\partial \tau_{ji}}{\partial x_j} + x_i$$

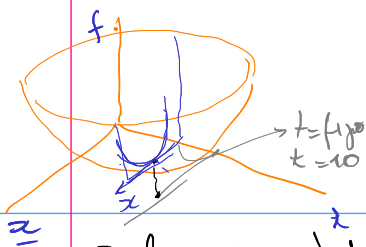
$\frac{d}{dt}$ $\rightarrow$ función escalar

$$\frac{D}{Dt} \left(\int_V \rho \cdot f \, dV \right) = \int_V \left[\frac{D}{Dt} (\rho \cdot f) + \rho \cdot f \cdot \frac{\partial v_j}{\partial x_j} \right] dV = \int_V \left(\frac{D}{Dt} \rho \cdot f + \rho \cdot \frac{Df}{Dt} + \rho \cdot f \cdot \frac{\partial v_j}{\partial x_j} \right) dV =$$

$$= \int_V f \left(\frac{D\rho}{Dt} + \rho \cdot \frac{\partial v_j}{\partial x_j} \right) dV + \int_V \rho \cdot \frac{Df}{Dt} dV = \int_V \rho \cdot \frac{Df}{Dt} dV$$

Por la ec. de continuidad
0

$$\frac{D}{Dt} \left(\int_V \rho \cdot f \, dV \right) = \int_V \rho \cdot \frac{Df}{Dt} dV$$



$$v = \frac{dx(a,t)}{dt} = \left. \frac{\partial x(a,t)}{\partial t} \right|_{a=f(t)}$$

$$f(x,t) = x_1 t$$

$$\frac{\partial f}{\partial x} = t$$

$$\frac{\partial f}{\partial t} = x_1$$

Balance de cantidad movimiento angular:

$$L_i = \int_V \epsilon_{ijk} x_j X_k dV + \int_S \epsilon_{ijk} x_j T_k dS$$

$$H_i = \int_V \epsilon_{ijk} x_j \rho v_k dV$$

$$\frac{D}{Dt} H_i = L_i$$

$$\frac{D}{Dt} H_i = \frac{D}{Dt} \int_V \epsilon_{ijk} x_j \rho v_k dV$$

$$= \int_V \left[\frac{\partial}{\partial t} (\epsilon_{ijk} x_j \rho v_k) + \frac{\partial}{\partial x_e} (\epsilon_{ijk} x_j \rho v_k v_e) \right] dV$$

$$\epsilon_{ijk} \left[\underbrace{\frac{\partial x_j}{\partial t} \rho v_k}_{v_j} + x_j \cdot \frac{\partial}{\partial t} (\rho v_k) \right]$$

$$\underbrace{\rho \epsilon_{ijk} v_j v_k}_0 + \epsilon_{ijk} x_j \cdot \frac{\partial}{\partial t} (\rho v_k)$$

$$= \int_V \left[\epsilon_{ijk} x_j \cdot \frac{\partial}{\partial t} (\rho v_k) + \frac{\partial}{\partial x_e} (\epsilon_{ijk} x_j \rho v_k v_e) \right] dV$$

$$+ \epsilon_{ijk} \frac{\partial x_j}{\partial x_e} (\rho v_k v_e) + \epsilon_{ijk} x_j \frac{\partial (\rho v_k v_e)}{\partial x_e}$$

$$+ \underbrace{\epsilon_{ijk} \rho v_k v_j}_0 + \epsilon_{ijk} x_j \frac{\partial (\rho v_k v_e)}{\partial x_j}$$

$$= \int_V \left[\epsilon_{ijk} x_j \frac{\partial}{\partial t} (\rho v_k) + \epsilon_{ijk} x_j \frac{\partial (\rho v_k v_e)}{\partial x_j} \right] dV$$

$$x(a,t)$$

$$\frac{Df(x,t)}{Dt} = \left. \frac{\partial f(x,t)}{\partial x} \right|_{t=f(t)} \cdot v + \left. \frac{\partial f(x,t)}{\partial t} \right|_{x=f(t)}$$

$$f = x(a,t) = x(t)$$

$$\frac{Df(x,t)}{Dt} = \frac{Dx(t)}{Dt} = \left. \frac{\partial x(t)}{\partial x} \right|_{t=f(t)} \cdot v + \left. \frac{\partial x(t)}{\partial t} \right|_{x=f(t)}$$

a aceleración de la partícula
a aceleración convectiva $\frac{dx}{dt} \rightarrow$ no es de cosa
a aceleración local

$$f(x,t) = v(x,t) \rightarrow \frac{Df(x,t)}{Dt} = \frac{\partial v(x,t)}{\partial x} \cdot v + \frac{\partial v(x,t)}{\partial t}$$

$$\frac{D}{Dt} \int_V A dV = \int_V \frac{\partial A}{\partial t} dV + \int_V \frac{\partial}{\partial x_j} (A v_j) dV$$

$$\epsilon_{ijk} v_j v_k = 0$$

$$\epsilon_{ijk} A_{jk} = 0 \text{ ya que } A_{jk} \text{ es simétrico}$$

$$\epsilon_{ijk} \frac{\partial A}{\partial x_j \partial x_k} = 0$$

$$L_i = \int_V \underbrace{e_{ijk} x_j X_k}_{\text{}} dV + \int_S e_{ijk} x_j T_k dS$$

Fórmula de Cauchy $T_k = n_x \tau_{ek}$

$$= \underbrace{\int_V e_{ijk} x_j X_k dV}_{\text{}} + \int_S e_{ijk} x_j \tau_{ek} n_e dS = \quad \left. \begin{array}{l} \text{Teorema de Gauss} \end{array} \right\}$$

$$= \underbrace{\int_V e_{ijk} x_j X_k dV}_{\text{}} + \int_V \frac{\partial}{\partial x_e} (e_{ijk} x_j \tau_{ek}) dV$$

$$= \underbrace{\int_V e_{ijk} x_j X_k dV}_{\text{}} + e_{ijk} \overset{\text{Der}}{\frac{\partial x_j}{\partial x_e}} \tau_{ek} + e_{ijk} x_j \frac{\partial \tau_{ek}}{\partial x_e} =$$

$$= \underbrace{\int_V e_{ijk} x_j X_k dV}_{\text{}} + e_{ijk} \tau_{jk} + e_{ijk} x_j \frac{\partial \tau_{ek}}{\partial x_e}$$

$$\frac{DH_i}{Dt} = L_i$$

$$\int_V \left[e_{ijk} x_j \cdot \frac{\partial}{\partial t} (\rho \cdot v_k) + e_{ijk} x_j \frac{\partial (\rho v_k v_e)}{\partial x_e} \right] dV = \int_V \left(e_{ijk} x_j X_k + e_{ijk} \tau_{jk} + e_{ijk} x_j \frac{\partial \tau_{ek}}{\partial x_e} \right) dV$$

$$\int_V e_{ijk} x_j \left(\frac{\partial}{\partial t} (\rho \cdot v_k) + \frac{\partial (\rho v_k v_e)}{\partial x_e} - X_k - \frac{\partial \tau_{ek}}{\partial x_e} \right) dV = \int_V e_{ijk} \tau_{jk} dV$$

$$\rightarrow \frac{\partial \rho}{\partial t} \cdot v_k + \rho \cdot \frac{\partial v_k}{\partial t} + \frac{\partial (\rho v_e)}{\partial x_e} \cdot v_k + \rho v_e \cdot \frac{\partial v_k}{\partial x_e} - X_k - \frac{\partial \tau_{ek}}{\partial x_e}$$

$$v_k \left(\frac{\partial \rho}{\partial t} + \frac{\partial (\rho v_e)}{\partial x_e} \right) + \rho \cdot \left(\frac{\partial v_k}{\partial t} + v_e \cdot \frac{\partial v_k}{\partial x_e} \right) - X_k - \frac{\partial \tau_{ek}}{\partial x_e}$$

Ec. de continuidad

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_j}{\partial x_j} = 0.$$

$$\rho \cdot \frac{Dv}{Dt} - X_k - \frac{\partial \tau_{ek}}{\partial x_e}$$

Por lo
Ec. de Movimiento
(o Balance de la
Cant. de Mov. lineal)

$$0 = \int_V e_{ijk} \tau_{jk} dV \quad \forall V$$

$$e_{ijk} \tau_{jk} = 0$$

$0 = e_{ijk} \Gamma_{jk} \rightarrow \alpha$ extiende para un i quinario:

$$e_{i11} \Gamma_{11} + e_{i12} \Gamma_{12} + e_{i13} \Gamma_{13} + e_{i21} \Gamma_{21} + e_{i22} \Gamma_{22} + e_{i23} \Gamma_{23} + e_{i31} \Gamma_{31} + e_{i32} \Gamma_{32} + e_{i33} \Gamma_{33} = 0$$

$$\text{Para } i=1 \Rightarrow e_{112} \Gamma_{12} + e_{113} \Gamma_{13} + e_{121} \Gamma_{21} + e_{123} \Gamma_{23} + e_{131} \Gamma_{31} + e_{132} \Gamma_{32} = 0$$

$$e_{123} \Gamma_{23} + e_{132} \Gamma_{32} = 0$$

$$\Gamma_{23} - \Gamma_{32} = 0 \Rightarrow \Gamma_{23} = \Gamma_{32}$$

Para $i=2 \Rightarrow \dots$

Para $i=3 \Rightarrow \dots$

$\Rightarrow$ El tensor $\underline{\Gamma}$ es simétrico

$$0 = e_{ijk} \Gamma_{jk} = \frac{1}{2} (e_{ijk} \Gamma_{jk} + e_{ijk} \Gamma_{jk}) = \frac{1}{2} (e_{ijk} \Gamma_{jk} - e_{ikj} \Gamma_{jk}) =$$

$\rightarrow$ Por índice dummy $\Rightarrow e_{ikj} \Gamma_{kj} = e_{ijk} \Gamma_{kj}$

$$= \frac{1}{2} (e_{ijk} \Gamma_{jk} - e_{ijk} \Gamma_{kj}) = \frac{1}{2} e_{ijk} (\Gamma_{jk} - \Gamma_{kj})$$

$$e_{ijk} (\Gamma_{jk} - \Gamma_{kj}) = 0$$

Para que α cumpla esta ecuación $\Rightarrow$

$$\Gamma_{jk} - \Gamma_{kj} = 0 \Rightarrow$$

$$\Gamma_{jk} = \Gamma_{kj} \Rightarrow \underline{\Gamma} \text{ es simétrico}$$